`genQE_3N` — Three-Nucleon GCF Event Generator

Physics & Pipeline Documentation

GCF with 3N Project

March 10, 2026

Contents

1 Overview

`genQE_3N` extends the two-nucleon GCF generator to **three-nucleon short-range correlations (3N SRC)**. It generates events of the form

$$e + A \longrightarrow e' + N_1 + N_2 + N_3 + (A{-}3)^*, \tag{1}$$

where the electron scatters off one nucleon of a correlated triplet. The current implementation targets ^{4}He with *ppn* triplets.

The key conceptual difference from the 2N generator is that *contact coefficients times universal functions* are replaced by **pre-computed 3D density matrices** $\rho(\theta_{ab}, k_{\rm cm}, k_{\rm rel})$ obtained from *ab initio* nuclear structure calculations.

2 From 2N Contacts to 3N Density Matrices

2.1 2N recap

In the 2N GCF the pair momentum distribution factorizes as

$$n^{(ij)}(\boldsymbol{k}) = C_{ij}^{A}\,|\varphi_{ij}(k)|^2, \tag{2}$$

separating a nucleus-dependent contact C_{ij}^{A} from a universal function φ_{ij}.

2.2 3N generalization

For three nucleons no analogous simple factorization exists. Instead, the full three-body correlation is encoded in a density matrix that depends on three independent kinematic variables describing the internal configuration of the triplet:

$$\rho_{3N}(\theta_{ab}, k_{\rm cm}, k_{\rm rel}), \tag{3}$$

where

- θ_{ab}: angle between the two Jacobi-like momenta $\boldsymbol{p}_a$ and $\boldsymbol{p}_b$ (see below),
- $k_{\rm cm}$: magnitude of the three-body center-of-mass–like momentum,
- $k_{\rm rel}$: magnitude of the relative momentum within the pair.

These density matrices are tabulated on a regular grid and loaded from binary data files at initialization.

2.3 Data files

Four NN-potential choices are available:

Index	File
1	`AV8_1D.dnst` (Argonne v_8')
2	`N2LO_1D.dnst` (Chiral N^{2}LO)
3	`G3_1D.dnst` (Chiral N^{3}LO / higher)
4	`AV4_1D.dnst` (Argonne AV4$'$)

Each file contains $61 \times 101 \times 101 = 622{,}261$ entries:

Variable	Bins	Range
θ_{ab}	61	$[0^\circ, 180^\circ]$, $\Delta\theta = 3^\circ$
$k_{\rm cm}$	101	$[0, 10]\,{\rm fm}^{-1}$, $\Delta k = 0.1\,{\rm fm}^{-1}$
$k_{\rm rel}$	101	$[0, 10]\,{\rm fm}^{-1}$, $\Delta k = 0.1\,{\rm fm}^{-1}$

Values are trilinearly interpolated at runtime.

3 Three-Body Kinematics

3.1 Coordinate system

The three nucleon momenta are parameterized via two Jacobi-like vectors $\boldsymbol{p}_a$, $\boldsymbol{p}_b$ and a three-body center-of-mass momentum $\boldsymbol{P}_{\rm CM}$.

Internal frame. $\boldsymbol{p}_a$ is placed along $\hat{z}$ and $\boldsymbol{p}_b$ lies in the xz-plane:

$$\boldsymbol{p}_a = (0,\, 0,\, p_a), \qquad \boldsymbol{p}_b = (p_b \sin\theta_{ab},\ 0,\ p_b \cos\theta_{ab}). \tag{4}$$

Rotation to lab. Three Euler angles $(\phi_a,\, \theta_a,\, \phi_{baz})$, sampled uniformly, rotate this frame into the laboratory.

Individual momenta.

$$\boldsymbol{v}_1 = \boldsymbol{p}_a, \qquad \boldsymbol{v}_2 = -\tfrac{1}{2}\,\boldsymbol{p}_a + \boldsymbol{p}_b, \qquad \boldsymbol{v}_3 = -\tfrac{1}{2}\,\boldsymbol{p}_a - \boldsymbol{p}_b. \tag{5}$$

After Euler rotation, each nucleon receives an additional $\boldsymbol{P}_{\rm CM}/3$ boost. The $(A-3)$ residual recoils with $\boldsymbol{p}_{A-3} = -\boldsymbol{P}_{\rm CM}$.

3.2 Center-of-mass motion

$$P_{{\rm CM},\alpha} \sim \mathcal{N}(0,\ \sigma_{\rm CM}), \qquad \sigma_{\rm CM} = 0.055\,{\rm GeV}/c. \tag{6}$$

4 Event Generation Pipeline

4.1 Step 1: Triplet type assignment

Currently only *ppn* triplets are generated. All three nucleons are initialized as protons, then exactly one — chosen uniformly at random among N_1, N_2, N_3 — is reassigned to a neutron. Thus the leading nucleon N_1 can be either a proton or a neutron (with probability 2/3 and 1/3 respectively). The weight carries a factor of 3 to account for the three equivalent assignments.

4.2 Step 2: Sample three-body CM momentum

$$\boldsymbol{P}_{\rm CM} \sim \mathcal{N}^3(0,\, \sigma_{\rm CM}). \tag{7}$$

4.3 Step 3: Sample orientation (Euler angles)

$$\phi_a \in [-\pi,\, \pi], \qquad \cos\theta_a \in [-1,\, 1], \qquad \phi_{baz} \in [-\pi,\, \pi]. \tag{8}$$

Phase-space weight factor: $\Delta\phi_a\,\Delta(\cos\theta_a)\,\Delta\phi_{baz} = 8\pi^2$.

4.4 Step 4: Sample relative momenta

$$p_a \in [0,\, 5.0]\,{\rm GeV}/c, \qquad p_b \in [0,\, 5.0]\,{\rm GeV}/c, \qquad \theta_{ab} \in [10^{-4},\, \pi]. \tag{9}$$

Phase-space weight factor: $5.0 \times 5.0 \times \pi$.

4.5 Step 5: Construct individual momenta

Build $\boldsymbol{v}_{1,2,3}$ from the Jacobi decomposition, apply the Euler rotation, add $\boldsymbol{P}_{\rm CM}/3$ to each, and set $\boldsymbol{p}_{A-3} = -\boldsymbol{P}_{\rm CM}$.

4.6 Step 6: Residual-nucleus energy

For ^{4}He the $(A{-}3) = {}^1$H system is a single nucleon:

$$m_{A-3} = m_N + E^*, \qquad E_{A-3} = \sqrt{|\boldsymbol{p}_{A-3}|^2 + m_{A-3}^2}. \tag{10}$$

Spectator energies:

$$E_2 = \sqrt{|\boldsymbol{v}_2|^2 + m_N^2}, \qquad E_3 = \sqrt{|\boldsymbol{v}_3|^2 + m_N^2}. \tag{11}$$

Leading nucleon energy from conservation:

$$E_1 = m_A - E_{A-3} - E_2 - E_3. \tag{12}$$

4.7 Step 7: Electron scattering vertex

The scattered electron direction is sampled uniformly:

$$\cos\theta_k \in [\cos 40^\circ,\, \cos 5^\circ], \qquad \phi_k \in [-\pi,\, \pi]. \tag{13}$$

Define the recoil system $\boldsymbol{X} = \boldsymbol{v}_2 + \boldsymbol{v}_3 + \boldsymbol{p}_{A-3}$ and $Z = m_A + E_{\text{beam}} - E_2 - E_3 - E_{A-3}$. The scattered electron energy is fixed by 4-momentum conservation:

$$E_k = \frac{Z^2 - m_N^2 - |\boldsymbol{X}|^2}{2\big(Z - \boldsymbol{X}\cdot\hat{k}\big)}. \tag{14}$$

The virtual photon is $\boldsymbol{q} = \boldsymbol{p}_{\text{beam}} - \boldsymbol{k}$, and the lead nucleon receives the photon: $\boldsymbol{p}_{\text{lead}} = \boldsymbol{v}_1 + \boldsymbol{q}$.

4.8 Step 8: Jacobian

The energy-conservation δ-function produces a Jacobian:

$$J_\delta = E_{\text{lead}} + \boldsymbol{p}_{\text{lead}} \cdot (\boldsymbol{v}_1 + \boldsymbol{p}_{\text{beam}} + \hat{k}), \qquad w \;\ast\!= \frac{E_{\text{lead}}}{J_\delta}. \tag{15}$$

4.9 Step 9: Cross section and density weight

$$w \;\ast\!= \frac{1}{2}\,(2\pi)^{-7}\,C\,t \;\times\; \sigma_{eN}(E_{\text{beam}},\, \boldsymbol{k},\, \boldsymbol{p}_{\text{lead}}) \;\times\; \rho_{3N}(\theta_{ab},\, p_a,\, p_b), \tag{16}$$

where $C = 0.12$ and $t = 2$ are normalization constants. σ_{eN} uses the same Kelly form factors as the 2N generator.

4.10 Step 10: Density matrix evaluation

The sampled variables $(p_a,\, p_b,\, \theta_{ab})$ must be transformed to the density-matrix coordinates $(k_{\text{cm}},\, k_{\text{rel}},\, \theta'_{ab})$ via a coordinate relabeling that depends on which pair (N_2, N_3) is the correlated subsystem. The Jacobian of this coordinate transformation is

$$J_{\text{coord}} = \frac{p_a\, p_b\, \sin\theta_{ab}}{p'_a\, p'_b\, \sin\theta'_{ab}}, \tag{17}$$

and the density is

$$\rho = J_{\text{coord}} \times \text{interpolate}\big[\rho_{\text{table}}(\theta'_{ab},\, k_{\text{cm}},\, k_{\text{rel}})\big]. \tag{18}$$

5 Weight Summary

Collecting all factors:

$$w = 3 \times 8\pi^2 \times 25\pi \times \Delta(\cos\theta_k)\,\Delta\phi_k \times \frac{E_{\text{lead}}}{J_\delta} \times \frac{C\,t}{2\,(2\pi)^7} \times \sigma_{eN} \times \rho_{3N} \times \Theta(w > 0). \tag{19}$$

6 Alternative Momentum Parameterization

For analysis purposes (Dalitz/ternary diagrams, spectral functions), an alternative parameterization in terms of momentum fractions is available:

$$f_i = \frac{|\boldsymbol{p}_i|}{|\boldsymbol{p}_1| + |\boldsymbol{p}_2| + |\boldsymbol{p}_3|}, \qquad f_1 + f_2 + f_3 = 1. \tag{20}$$

The Jacobian for the transformation $(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) \to (p_{\text{tot}}, f_1, f_2)$ is

$$J = \frac{8}{\sqrt{3}}\,\frac{1}{p_1\,p_2\,p_3\,p_{\text{tot}}^2}. \tag{21}$$

7 Differences from the 2N Generator

Aspect	`genQE` (2N)	`genQE_3N` (3N)
Correlated cluster	Pair (NN)	Triplet (NNN)
Structure input	Contacts $C_{ij}\,\varphi^2(k)$	3D density matrix $\rho(\theta, k_{\text{cm}}, k_{\text{rel}})$
Residual system	$(A{-}2)^*$	$(A{-}3)^*$
Phase space dim.	6D (CM + rel)	9D (CM + Euler + Jacobi)
Electron vertex	Quadratic solve for $k_\perp$	Direct solve for E_k
σ_{CM}	0.10–0.15 GeV/c	0.055 GeV/c
Target nuclei	^{2}H through ^{208}Pb	^{4}He

8 Summary of Algorithm

For each Monte Carlo trial:

1. Assign triplet types (ppn); weight $\times 3$.
2. Sample $\boldsymbol{P}_{\text{CM}}$ (Gaussian, $\sigma = 0.055\,\text{GeV}/c$).
3. Sample Euler angles $(\phi_a, \theta_a, \phi_{baz})$ uniformly.
4. Sample Jacobi magnitudes (p_a, p_b) and angle θ_{ab} uniformly.
5. Construct $\boldsymbol{v}_{1,2,3}$ via Jacobi decomposition + Euler rotation + CM boost.
6. Compute residual and spectator energies; derive E_1 from conservation.
7. Sample electron direction (θ_k, ϕ_k); solve for E_k.
8. Compute Jacobian J_δ from energy-conservation constraint.
9. Evaluate σ_{eN} (CC1, Kelly form factors).
10. Look up ρ_{3N} from density matrix with coordinate transform.
11. Multiply all weight factors; store event if $w > 0$.